

Global Food Security And Agricultural Market Analytics

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1. Objective

The objective of this project is to analyze the relationship between **commodity prices, fertilizer prices, crop production, and food security indicators** in order to identify trends, risks, and possible food security challenges over time.

The dashboard helps answer questions such as:

- Are food prices increasing?
- Are fertilizer costs affecting agriculture?
- Is crop production improving or declining?
- Are crisis and supply stress situations rising?
- How are agricultural inputs linked to food security outcomes?

2. Aim of the Project

The aim of this project is to build an interactive Power BI dashboard that gives a complete view of:

- commodity price movement,
- agricultural productivity,
- food affordability,
- food crisis indicators,
- and long-term food security risk patterns.

This project supports decision-making for analysts, policymakers, and business stakeholders by turning raw data into meaningful insights.

3. Project Requirement

This project is built using four datasets:

A. Commodity Prices

Contains:

- Wheat price
- Maize price
- Soybean oil price
- DAP fertilizer price
- MOP potash price
- month-over-month and year-over-year growth metrics
- food price index indicators

B. Crop Yields

Contains:

- maize area and production
- rice paddy area and production
- wheat area and production
- total staple production

C. Food Security Panel

Contains:

- FFPI overall
- FFPI cereals
- FFPI above baseline
- crisis flag
- supply stress flag
- food shock score
- cereals affordability index
- volatility indicators
- date and year fields

D. Food Security Indicators

Contains:

- undernourishment percentage

- food production index
- cereal yield
- arable land percentage
- growth indicators

4. Project Analysis

This dashboard is designed into **five pages**, each page focusing on a different business question.

Page 1: Executive Summary

This page gives a high-level overview of the entire project.

Charts used

- KPI cards for major growth indicators
- Food price trend line chart
- Commodity price comparison chart
- Fertilizer price trend chart

What it shows

- Overall movement of food prices over time
- Comparison between wheat, maize, and soybean oil prices
- Fertilizer cost trend and its possible effect on agriculture
- Quick summary of the current market and food security situation

Purpose

This page helps users understand the big picture in a single view.

Page 2: Commodity Price Analysis

This page focuses on price behavior of major commodities and fertilizers.

Charts used

- Commodity price trend line chart
- Month-over-month growth chart for wheat, maize, and soybean oil
- Fertilizer price trend chart
- Average price KPI cards

What it shows

- Price fluctuations across time
- Short-term growth and decline patterns
- Which commodity is more volatile
- Whether fertilizer prices are rising faster than crop prices

Purpose

This page helps identify inflationary pressure in food and farm input markets.

Page 3: Crop Yield & Agriculture Analysis

This page studies agricultural output and crop performance.

Charts used

- Food production index share donut chart
- Crop production comparison chart
- Cereal yield trend chart
- Undernourishment table
- KPI cards for total crop production and yield measures

What it shows

- How much maize, rice, and wheat are being produced

- Whether cereal yield is improving over time
- How food production index changes across years
- Relationship between production and undernourishment

Purpose

This page helps assess whether agriculture is producing enough food to meet demand.

Page 4: Food Security & Risk Analysis

This page highlights food security risk, affordability, and crisis signals.

Charts used

- Affordability analysis line chart
- Crisis and supply stress by year treemap
- Risk correlation scatter chart
- Food shock analysis line chart
- KPI cards for supply stress, crisis count, food production index, and FFPI

What it shows

- Whether cereals are becoming more or less affordable
- Years with higher crisis and supply stress events
- Relationship between FFPI and food shock score
- Food system instability over time

Purpose

This page identifies risk periods and helps detect warning signs of food insecurity.

Page 5: Detailed Insights

This page gives deeper analytical understanding of the drivers behind food insecurity.

Charts used

- FFPI above baseline trend line
- Crop yield and fertilizer application trend chart
- Agricultural land use vs food production index chart
- KPI cards for average arable land, cereal yield, food shock score, and FFPI above baseline

What it shows

- How often FFPI rises above baseline
- Whether fertilizer usage supports better yield
- Whether arable land and production index move together
- Root-cause style interpretation of food risk

Purpose

This page explains the underlying factors influencing food security outcomes.

5. Tools Required

The following tools were used for this project:

- **Microsoft Excel / CSV files** for source data preparation
- **Jupyter Notebook / Python** for data cleaning and preprocessing
- **Pandas** for data handling
- **Power BI** for dashboard creation
- **DAX** for calculated measures and KPI logic
- **Data Modeling** for relationships between tables

Data Preprocessing and Feature Engineering Using Python

Introduction

Before developing the Power BI dashboard, the raw datasets underwent extensive preprocessing using Python in Jupyter Notebook. The purpose of this phase was to ensure data quality, consistency, and readiness for analytical reporting.

The preprocessing stage focused on:

- Data cleaning
- Missing value treatment
- Date formatting
- Data type conversion
- Feature engineering
- Calculation of growth indicators
- Exporting cleaned datasets for Power BI integration

Python Environment Used

Libraries Used

```
import pandas as pd
import numpy as np
```

Tool Used

- Jupyter Notebook
- Python
- Pandas
- NumPy

Data Cleaning Process

1. Loading CSV Files

The raw CSV datasets were imported into Python using Pandas.

```
commodity_prices = pd.read_csv("commodity_prices.csv")
crop_yields = pd.read_csv("crop_yields.csv")
food_security_panel = pd.read_csv("food_security_panel.csv")
food_security_indicators = pd.read_csv("food_security_indicators.csv")
```

Purpose

To load structured data into DataFrames for further processing.

2. Checking Missing Values

Missing values were identified using:

```
commodity_prices.isnull().sum()
```

Purpose

To identify incomplete records that could affect analysis.

3. Handling Missing Values

Numerical columns were imputed using mean values where necessary.

```
commodity_prices.fillna(
    commodity_prices.mean(numeric_only=True),
    inplace=True
)
```

Purpose

To avoid blank values in Power BI visuals and calculations.

4. Date Conversion

Date fields were converted into datetime format.

```
commodity_prices['date'] = pd.to_datetime(  
    commodity_prices['date']  
)
```

Purpose

To support time-series analysis and growth calculations.

5. Creating Year and Month Columns

Additional date attributes were extracted.

```
commodity_prices['Year'] = commodity_prices['date'].dt.year
```

```
commodity_prices['Month'] = commodity_prices['date'].dt.month
```

Purpose

To simplify filtering and trend analysis in Power BI.

6. Duplicate Record Removal

```
commodity_prices.drop_duplicates(inplace=True)
```

Purpose

To prevent duplicate records from inflating calculations.

Feature Engineering Using Python

Feature engineering was performed to create additional analytical indicators required for dashboard development.

1. Month-over-Month Growth Calculation

Wheat

```
commodity_prices['wheat_mom_growth_pct'] = (  
    commodity_prices['wheat_usd_mt']  
    .pct_change()  
) * 100
```

Maize

```
commodity_prices['maize_mom_growth_pct'] = (  
    commodity_prices['maize_usd_mt']  
    .pct_change()  
) * 100
```

Soybean Oil

```
commodity_prices['soybean_mom_growth_pct'] = (  
    commodity_prices['soybean_oil_usd_mt']  
    .pct_change()  
) * 100
```

Purpose

To measure short-term commodity price changes.

2. Year-over-Year Growth Calculation

Wheat

```
commodity_prices['wheat_yoy_growth_pct'] = (  
    commodity_prices['wheat_usd_mt']  
    .pct_change( periods=12 )  
    ) * 100
```

Purpose

To measure annual price movement and long-term inflation trends.

3. Average Commodity Price Indicators

Python was used to validate average values.

```
avg_wheat = commodity_prices[  
    'wheat_usd_mt'  
].mean()
```

```
avg_maize = commodity_prices[  
    'maize_usd_mt'  
].mean()
```

Purpose

To verify Power BI DAX calculations.

4. Crisis and Supply Stress Analysis

```
crisis_count = food_security_panel[  
    'ffpi_crisis_flag'  
].sum()
```

```
supply_stress_count = food_security_panel[  
    'supply_stress_flag'  
].sum()
```

Purpose

To quantify food security risk events.

Exporting Cleaned Data

After preprocessing, the datasets were exported as cleaned CSV files.

```
commodity_prices.to_csv(  
    "commodity_prices_cleaned.csv",  
    index=False  
)
```

Purpose

To create a structured data source for Power BI.

Benefits of Python Preprocessing

Using Python provided several advantages:

Improved Data Quality

- Missing values handled
- Duplicate records removed
- Standardized formats

Faster Analysis

- Pre-calculated growth indicators
- Reduced Power BI transformation workload

Better Dashboard Performance

- Smaller and optimized datasets
- Faster report refresh times

Accurate Reporting

- Consistent calculations between Python and Power BI

6. DAX Used

Below are the important DAX measures used in the project.

A. Average Commodity Price Measures

Average Wheat Price = AVERAGE(commodity_prices[wheat_usd_mt])

Average Maize Price = AVERAGE(commodity_prices[maize_usd_mt])

Average Soybean Oil Price = AVERAGE(commodity_prices[soybean_oil_usd_mt])

Average DAP Fertilizer Price = AVERAGE(commodity_prices[dap_fertiliser_usd_mt])

Average MOP Potash Price = AVERAGE(commodity_prices[mop_potash_usd_mt])

Purpose

These measures are used in KPI cards and price trend visuals.

B. Food Price Index Measures

Average FFPI =

AVERAGE(food_security_panel[ffpi_overall])

Average FFPI Cereals =

AVERAGE(food_security_panel[ffpi_cereals])

Purpose

These measures track overall food inflation and cereal price pressure.

C. Growth Measures

Wheat MoM Growth % =

DIVIDE(
 [Average Wheat Price] - [Previous Month Wheat Price],
 [Previous Month Wheat Price]
) * 100

Maize MoM Growth % =

DIVIDE(
 [Average Maize Price] - [Previous Month Maize Price],
 [Previous Month Maize Price]
) * 100

Soybean Oil MoM Growth % =

DIVIDE(
 [Average Soybean Oil Price] - [Previous Month Soybean Oil Price],
 [Previous Month Soybean Oil Price]
) * 100

Wheat YoY Growth % =

DIVIDE(
 [Average Wheat Price] - [Previous Year Wheat Price],
 [Previous Year Wheat Price]
) * 100

Purpose

These measures show short-term and long-term price movement.

D. Crisis and Supply Stress Measures

If the flags are stored as **1** and **0**, then:

Crisis Count = SUM(food_security_panel[ffpi_crisis_flag])

Supply Stress Count = SUM(food_security_panel[supply_stress_flag])

Purpose

These measures count how many times crisis or supply stress occurred.

E. Agricultural Indicator Measures

Average Crop Yield =

AVERAGE(food_security_indicators[cereal_yield_kg_ha])

Average Arable Land =

AVERAGE(food_security_indicators[arable_land_pct])

Average Undernourishment =

AVERAGE(food_security_indicators[undernourishment_pct])

Average Food Production Index =

AVERAGE(food_security_indicators[food_production_index])

Purpose

These measures are used in agriculture and food security analysis.

F. Status Measures for Better Visuals

Crisis Status =

IF(

food_security_panel[ffpi_crisis_flag] = 1,


```
"Crisis",  
"No Crisis"  
)  
Supply Stress Status =  
IF(  
  food_security_panel[supply_stress_flag] = 1,  
  "Supply Stress",  
  "Normal"  
)  
FFPI Above Baseline Status =  
IF(  
  food_security_panel[ffpi_above_baseline] = 1,  
  "Above Baseline",  
  "Normal"  
)
```

Purpose

These are useful for donut charts, treemaps, and clearer reporting.

G. Target Measure for Gauge

Undernourishment Target = 5

Purpose

Used as a benchmark in the gauge chart to compare actual undernourishment against a target value.

7. Chart Explanation in Detail

Executive Summary Charts

1. Food Price Trend Line Chart

This chart shows how overall food price index changes over time. It helps identify inflation spikes and market instability.

2. Commodity Price Comparison Chart

This chart compares wheat, maize, and soybean oil prices together. It helps identify which commodity is more expensive or more volatile.

3. Fertilizer Price Trend Chart

This chart shows fertilizer price movement across time. It is important because fertilizer cost directly affects agricultural production.

4. KPI Cards

These cards provide quick numeric summaries such as average prices and growth values for fast decision-making.

Commodity Price Analysis Charts

1. Commodity Trend Line Chart

Shows the long-term movement of wheat, maize, and soybean oil prices.

2. MoM Growth Charts

These charts show month-to-month price increase or decrease. They are useful for spotting sudden inflation or deflation.

3. Average Price Cards

These cards summarize the average market price for each commodity.

Crop Yield & Agriculture Analysis Charts

1. Donut Chart for Food Production Index Share

Shows the share of production index across years or categories.

2. Crop Production Comparison

Compares production of maize, wheat, and rice paddy across years.

3. Cereal Yield Trend

Shows whether yield per hectare is improving or declining.

4. Undernourishment Table

Provides a detailed yearly view of undernourishment and production indicators.

Food Security & Risk Analysis Charts

1. Affordability Analysis

Shows whether cereals are becoming more or less affordable over time.

2. Treemap of Crisis and Supply Stress

Displays the concentration of crisis and supply stress events by year.

3. Risk Correlation Scatter Chart

This chart analyzes the relationship between food price levels and food shock score.

4. Food Shock Analysis

Shows periods when the food system faced higher stress or disruption.

Detailed Insights Charts

1. FFPI Above Baseline Trend

Shows periods when food prices moved above normal baseline levels.

2. Crop Yield and Fertilizer Application Trend

Helps understand whether fertilizer use supports better crop yield.

3. Agricultural Land Use vs Food Production Index

Shows how land availability relates to food production performance.

4. KPI Cards

Summarize major agriculture and food security indicators.

8. Project Conclusion

This project successfully combines commodity prices, crop production, fertilizer costs, and food security indicators into one integrated Power BI dashboard.

The analysis shows:

- how food and fertilizer prices change over time,
- how crop production responds to these changes,
- how undernourishment and food shock risk behave,
- and how crisis and supply stress indicators reveal food security pressure.

This dashboard provides a complete analytical view of the food supply chain, helping identify the relationship between commodity inflation, agricultural productivity, and food security risk. It supports informed decision-making by highlighting trends, risk signals, and production challenges in a clear visual format.